DEPARTMENT OF MATHEMATICS, IIT MADRAS

**MA5950 — Mathematical Finance: Theory & Implementation**

Instructor: Prof. Neelesh Upadhye

# Lecture 12: Beta Estimation & Extensions

The Market Model · Rolling Beta · Critiques of CAPM

August 13, 2026

# Today's Agenda

1. The market model: estimating beta by regression

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2. $R^2$: how much risk is systematic?
3. Rolling beta
4. Critiques and extensions of CAPM

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- Interpret $R^2$ as the fraction of risk that is systematic.
- Explain why beta is estimated on a rolling basis.
- Describe CAPM's main empirical critiques and extensions.

# Beta Is Estimated, Not Observed

## The Market Model / Characteristic Line

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}$$

# Beta Is Estimated, Not Observed

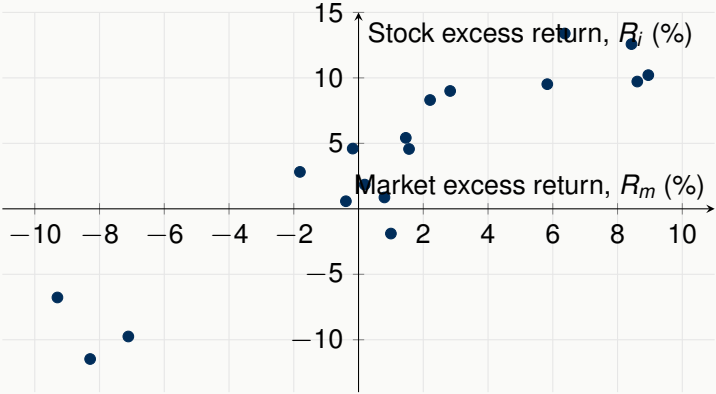
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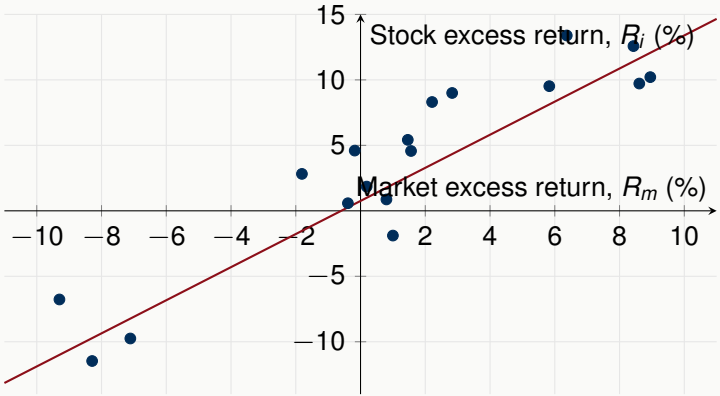
## Remark

OLS gives $\hat{\beta}_i = \widehat{\mathrm{Cov}}(R_i, R_m) / \widehat{\mathrm{Var}}(R_m)$ – same formula as Lecture 10, now estimated from a sample.

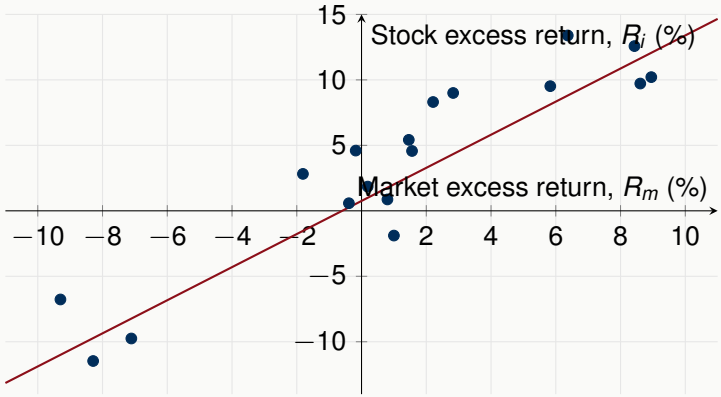
# The Characteristic Line, Fitted



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$\hat{\beta} \approx 1.26$, $\hat{\alpha} \approx 0.75\%$/month. Scatter around the line = unsystematic risk.

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**The Risk Pie**

$R^2$ = the systematic slice, moving with the market. $1 - R^2$ = the unsystematic slice, firm-specific noise. $R^2 = 0.82$ (our scatter, $\rho \approx 0.91$): 82% of this stock's variance is market-driven.

# Why Re-Estimate Beta Repeatedly?

**Remark**

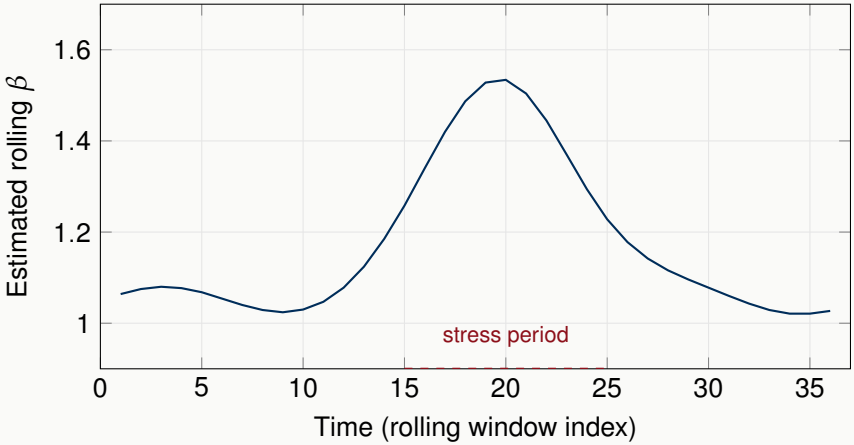
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**Remark**

Business mix, leverage, and the economy all change over time. No reason $\beta_i$ stays constant – analysts re-estimate on a rolling window (e.g., trailing 24 months), re-run every month.

# Rolling Beta Through a Stress Period



# Why Beta Spikes in a Crisis

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# Why Beta Spikes in a Crisis

**A Shared Shock Swamps Firm-Specific Noise**

Calm times: fortunes driven by products, management, customers – large unsystematic slice. Systemic crisis: nearly everyone hit by the *same* macro shock – correlations rise, firm stories stop mattering. Rolling beta spikes: a direct, measurable symptom.

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**Systematic risk**: undiversifiable, market-wide component of *one* asset's risk – measured by beta. **Systemic risk**: distress at one institution cascading to threaten the *entire* financial system. Rolling beta is a useful *tool* for monitoring systemic risk build-up – but the terms are not synonyms.

# Critiques and Extensions of CAPM

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- **Low-beta anomaly**: low-beta stocks historically earn *higher* risk-adjusted returns than CAPM predicts; high-beta, lower.
- **Multi-factor models**: Fama–French and descendants add size and value/growth factors – CAPM is the starting point, not the final word.

# Key Takeaways

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- **Rolling beta** tracks changing systematic exposure – often rising during systemic stress.
- CAPM has real critiques (Roll, low-beta anomaly) and has been extended by multi-factor models.

# Check Your Understanding

## Discuss (2 minutes)

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2. Why is using the Nifty 50 as "the market" not entirely innocuous (Roll's critique)?
3. If rolling beta rises sharply, what's happening to the systematic/unsystematic split?

# Looking Ahead

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# Module 3: Discrete-Time Pricing – Week 5

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# Module 3: Discrete-Time Pricing – Week 5

The one-period binomial model and the no-arbitrage principle.